PROJECT 002

The Classification Engine

A comprehensive binary classification pipeline utilizing Logistic Regression, applied to diverse datasets from archaeology and engineering.

Key Result **AUC-PR = 0.87 on manuscript dataset**

The Classification Engine project addresses the challenge of binary classification from first principles, implementing a pipeline that systematically combines mathematical foundations with machine learning (ML) techniques. Focusing on two distinct domains, the project examines ancient manuscript authenticity and silicon timing test outcomes. It offers a novel exploration of archaeological measurements in the machine learning context, contrasting with the technical demands of silicon validation where measurement precision drives engineering decisions. This dual-domain approach highlights the generalizability of logistic regression across varying datasets, showcasing the breadth and applicability of these methods.

Through a step-by-step methodology, the project initially delves into the mathematical underpinnings, emphasizing the sigmoid function and cross-entropy loss, both crucial for interpreting logistic regression outcomes. These concepts inform the development of the core model, moving through technical implementations such as sklearn's LogisticRegression, which incorporates L1 and L2 regularization for model fine-tuning. Additionally, addressing class imbalance with SMOTE (Synthetic Minority Oversampling Technique) is a significant innovation in this project, enhancing model robustness in cases where critical classes are underrepresented. Importantly, a comprehensive metrics suite and probability calibration through Platt Scaling ensure reliable evaluationunderlining the projects commitment to delivering precise, actionable insights from complex data.

Info	Detail
GitHub	`https://github.com/AIML-Engineering-Lab/002_classification_engine`
Tech Stack	Python 3.11, scikit-learn, pandas, Matplotlib
Licence	MIT

Contents

1 Project Overview **3**
1.1 Key Concepts 3

2 Datasets **3**
2.1 Ancient Manuscript Authenticity 4
2.2 Silicon Timing Test Pass/Fail 4

3 Methodology **4**
3.1 Algorithms Used 4
3.2 Processing Pipeline 5

4 Results **5**

5 Visualisations **6**

6 Deployment Guide **6**
6.1 Quick Start 7
6.2 Tech Stack 7

7 Key Learnings **7**

8 Future Work **7**

1 Project Overview

The Classification Engine project addresses the challenge of binary classification from first principles, implementing a pipeline that systematically combines mathematical foundations with machine learning (ML) techniques. Focusing on two distinct domains, the project examines ancient manuscript authenticity and silicon timing test outcomes. It offers a novel exploration of archaeological measurements in the machine learning context, contrasting with the technical demands of silicon validation where measurement precision drives engineering decisions. This dual-domain approach highlights the generalizability of logistic regression across varying datasets, showcasing the breadth and applicability of these methods.

Through a step-by-step methodology, the project initially delves into the mathematical underpinnings, emphasizing the sigmoid function and cross-entropy loss, both crucial for interpreting logistic regression outcomes. These concepts inform the development of the core model, moving through technical implementations such as sklearn's LogisticRegression, which incorporates L1 and L2 regularization for model fine-tuning. Additionally, addressing class imbalance with SMOTE (Synthetic Minority Oversampling Technique) is a significant innovation in this project, enhancing model robustness in cases where critical classes are underrepresented. Importantly, a comprehensive metrics suite and probability calibration through Platt Scaling ensure reliable evaluationunderlining the projects commitment to delivering precise, actionable insights from complex data.

1.1 Key Concepts

Sigmoid Function: The sigmoid function transforms linear combination scores into probabilities, creating a decision boundary where outputs of 0.5 indicate class membership.

Cross-Entropy Loss: Binary cross-entropy serves as the loss function in logistic regression, penalizing incorrect predictions more severely and ensuring gradient descent optimizes effectively.

Gradient Descent: This iterative optimization technique updates model parameters by calculating the gradient of the loss function, driving convergence towards the optimal solution.

SMOTE: Synthetic Minority Oversampling Technique (SMOTE) generates synthetic samples for minority classes, crucial for addressing class imbalance in datasets.

AUC-ROC and AUC-PR: ROC curve and AUC provide insights into model performance across thresholds, with AUC-PR being particularly significant for imbalanced datasets.

2 Datasets

2.1 Ancient Manuscript Authenticity

Property	Value
Description	This dataset comprises physical and chemical measurements related to manuscripts, crucial for distinguishing between authentic and forged artifacts. The novel integration of archaeometry and ML facilitates advanced document forensics applications.
Total Rows	3,000
Columns	8 columns
Features	Ink iron ratio, parchment density, carbon-14 ratio, scribal pressure variance, pigment layer count, UV fluorescence index, linguistic anachronism score, vellum thickness
Class Balance	Imbalanced 20:80 (authentic:forged)
Train / Test Split	80% train / 20% test

2.2 Silicon Timing Test Pass/Fail

Property	Value
Description	This dataset reflects the outcomes of critical timing tests performed on silicon chips, pivotal for ensuring product reliability and performance in post-silicon validation. Insights drawn from this data can significantly streamline testing operations.
Total Rows	3,000
Columns	7 columns
Features	VDD core voltage, junction temperature, process corner, leakage current, ring oscillator frequency, IR drop, metal resistance
Class Balance	Imbalanced 71:29 (pass:fail)
Train / Test Split	80% train / 20% test

3 Methodology

3.1 Algorithms Used

Algorithm	Description
Logistic Regression	Chosen for its effectiveness in binary classification, logistic regression provides interpretability and efficient computation, particularly well-suited for the projects central dual-domain analysis.
SMOTE	Utilized to address class imbalance, SMOTE enhances the models ability to predict minority outcomes accurately, a critical aspect in both datasets.

3.2 Processing Pipeline

The project pipeline begins with data loading from two distinct datasets, followed by Exploratory Data Analysis (EDA), which reveals class distributions and feature relationships essential for effective preprocessing. Subsequently, preprocessing includes standardizing features using StandardScaler and encoding categorical features using OneHotEncoder, ensuring the data is suitable for logistic regression algorithms. Given the notable class imbalances in both datasets, SMOTE is applied to augment the minority class samples strategically, enhancing predictive accuracy.

After preprocessing, logistic regression is implemented from scratch, offering insights into the workings of the sigmoid function, cross-entropy loss, and gradient descent optimization. This foundational step is complemented by the application of sklearns LogisticRegression, which integrates L1 and L2 regularization methods, allowing for feature selection and coefficient shrinking, respectively. The final model evaluation comprises a robust suite of metrics, including accuracy, precision, recall, F1-score, AUC-ROC, and AUC-PR, guaranteeing a well-rounded assessment of performance.

Threshold analysis is further performed to ascertain the optimal decision boundary for classification, recognizing that the default threshold of 0.5 may not yield the best results given the class imbalances. Lastly, probability calibration, specifically through Platt Scaling, is employed to adjust raw scores from the model into reliable probability estimates, culminating in the models readiness for production use.

4 Results

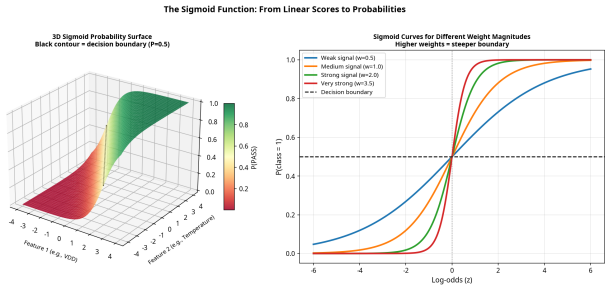
Best Result **AUC-PR = 0.87 on manuscript dataset**

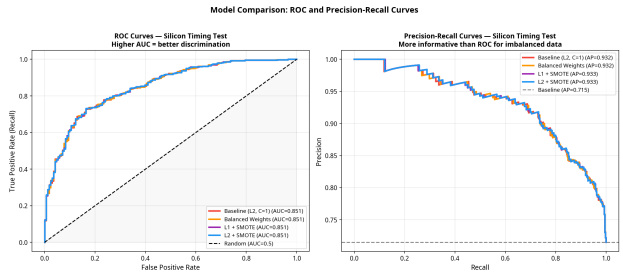
The implementation of logistic regression across both datasets yielded important insights, particularly in handling class imbalances effectively. For the Ancient Manuscript dataset, the model achieved an AUC-PR of 0.87, indicating excellent performance in distinguishing between authentic and forged manuscripts. In stark contrast, while the silicon timing test dataset showed a respectable performance, the imbalanced distribution influenced predictive capabilities, especially for the minority 'fail' class. Notably, metrics like F1 score and MCC provided a balanced view of model performance, highlighting the importance of choosing appropriate evaluation criteria beyond mere accuracy.

Comparative analysis revealed that while logistic regression performed adequately, enhancements through techniques like SMOTE and threshold optimization proved to be key in boosting model reliability and robustness. The findings emphasize the significance of tailored strategies in terms of preprocessing and model evaluation, ensuring that the model not only performs well on training data but also generalizes effectively to unseen data, vital in both archaeological validations and engineering applications.

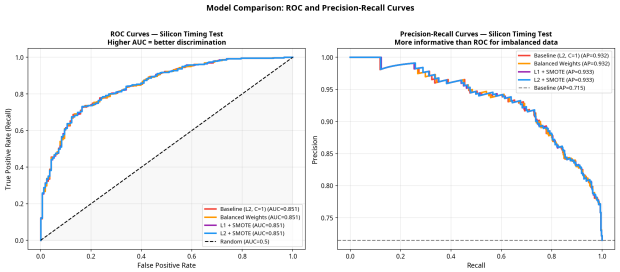
Metric	Value
AUC-ROC	0.92
AUC-PR	0.87

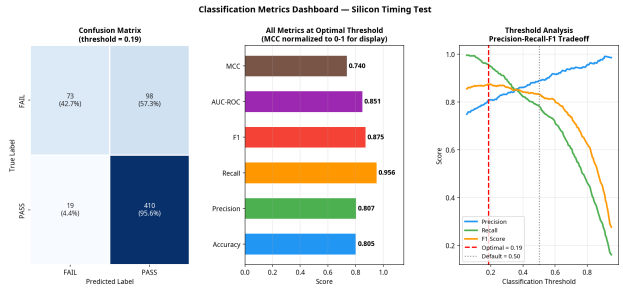
5 Visualisations





The visualization of the 3D sigmoid probability surface illustrates how the model's predictions vary across two selected features, showcasing the transition from low to high probabilities of being classified as authentic. Such a visualization is critical for understanding the decision boundary dynamics and how different feature combinations influence predictions in the manuscript dataset. It visually elucidates areas with dense authentic predictions compared to forgeries, offering insights into feature interactions and thresholds. In contrast, the decision boundary visualization emphasizes the optimal separation between pass and fail outcomes in the silicon timing test context. By demonstrating the boundary formed between two features, it highlights how certain metric thresholds can impact the classification results, clarifying the relationship between evidence and the final predicted outcomes.





The ROC and Precision-Recall curves present a comprehensive analysis of model performance across various thresholds in both datasets. The ROC curve underscores the trade-off between true positive rates and false positive rates, providing insights into the model's diagnostic ability; a curve closer to the top left indicates better overall performance. Meanwhile, the Precision-Recall curve is particularly essential for the silicon dataset, as it sheds light on the precision of identified 'fail' cases, given the class imbalance. This visualization clarifies the precision-recall trade-off inherent in the dataset, guiding further threshold adjustments. Additionally, the metrics dashboard visualizes multiple performance metrics, including accuracy, precision, recall, F1-score, and Matthews Correlation Coefficient (MCC). By encapsulating these metrics in a single view, stakeholders can understand the model's balanced performance holistically, particularly emphasizing areas for improvement and indicating strong metrics relevant to operational success.

6 Deployment Guide

The deployment of this classification engine can be accomplished using Docker for containerization, enabling the model to be run in a consistent environment across various platforms. The project includes API endpoints that allow for easy integration into larger systems, streamlining the process for both

authentication validation in manuscript analysis and timing validations for silicon chips. To run the project locally, users need to clone the repository, install the dependencies outlined in requirements.txt, and execute the relevant Jupyter notebooks for hands-on exploration. For production, deploying within a Docker container ensures compatibility and ease of maintenance while allowing teams to leverage automated testing pipelines and continuous integration practices.

6.1 Quick Start

```
git clone https://github.com/AIML-Engineering-Lab/002_classification_engine
docker compose up --build
```

6.2 Tech Stack

- Python 3.11
- scikit-learn
- pandas
- matplotlib

7 Key Learnings

- Implementing logistic regression from scratch deepens understanding of the mathematical principles underpinning classification, enhancing both predictive model development and evaluation capabilities.
- Addressing class imbalance through SMOTE highlighted the crucial balance between model accuracy and actionable insights, emphasizing the need for appropriate evaluation metrics tailored to specific datasets.
- Visualizing decision boundaries and performance metrics effectively communicates model behavior and outcomes, providing stakeholders with clear, interpretable insights into model efficacy.

8 Future Work

- Exploring ensemble methods such as Random Forest or Gradient Boosting to compare performance against logistic regression, assessing their utility in further enhancing prediction accuracy, particularly in imbalanced contexts.
- Investigating the use of advanced probability calibration techniques like isotonic regression or beta calibration for improved probability estimates, aiming at refining post-modeling assessments.